# Basic Networking Skills for Digital Forensics

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1. Executive Conclusion

Browser history records show visits to pages associated with product listings, email services, cryptocurrency-related resources, and downloaded files that appear consistent with payment documentation. The chronological sequence suggests a workflow involving advertisement viewing or management, communication activity, payment verification, and file retention. However, the Chrome History database records browser activity only and does not independently prove the content viewed, the completion of any transaction, the identity of the operator, or the actual exchange of controlled substances.

2. Evidence Acquisition and Integrity

| Field | Value |
|---|---|
| Original file name | History |
| Source | Assessment Case Materials |
| File type | Chrome History file |
| SHHA-256 (original) | b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 |
| SHA - 256 (working copy) | b991b0fa69662c44caf599e468c8088576ebd9cebbf46d02c89cf51bcde3fd19 |
| Analysis System | Kali Linux |
| Note | Original Copy not tempered |

2.1 Case folder Structure

```
┌──(root㉿kali)-[~]
└─# mkdir -p ~/SBT-DF204-Case2/{evidence,working,exported,reports,screenshots}
```

*Image 1 (EO1). mkdir -p ~/SBT-DF204-Case2/{evidence,working,exported, reports and screenshots} – creates the subfolders for evidence and file storage throughout the studies.*

## 2.2 Downloading evidence file

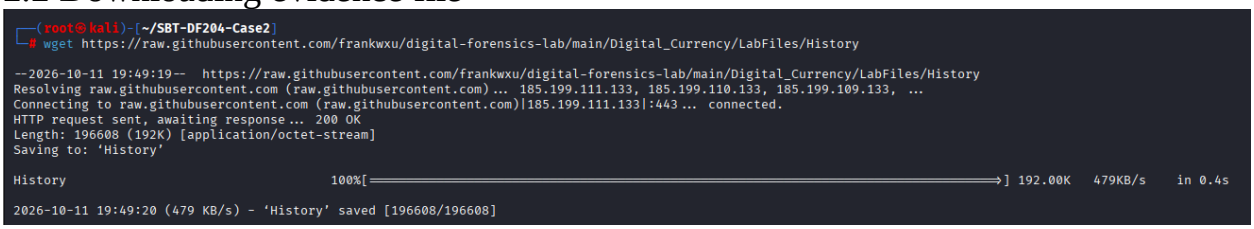



*Image 2 (EO2). The file History was successfully downloaded and saved*

## 2.3 Verifying Integrity

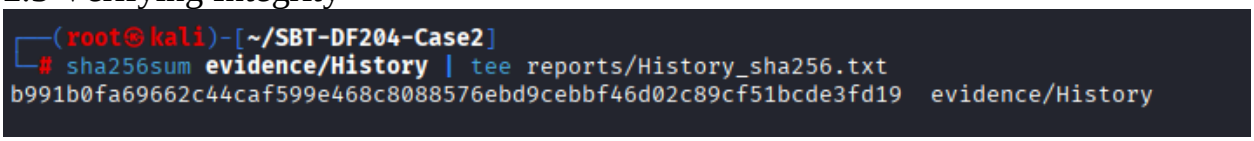



*Image 3 (EO3). SHA256sum evidence/History | tee reports/History_Sha256.txt, recorded the SHA256 hash of the original evidence file*

## 2.4 Creating a working Path

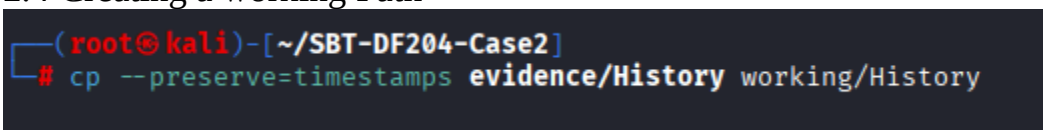



*Image 4 (E04). Cp –preserve=timestamps evidence/History working/History. A working part is created to preserve integrity*

3. Examining the file (History)

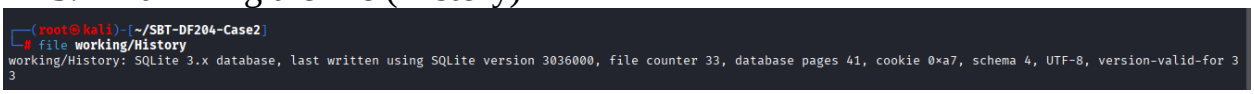



*Image 5 (E05). File working/History, checking the file type and tables*

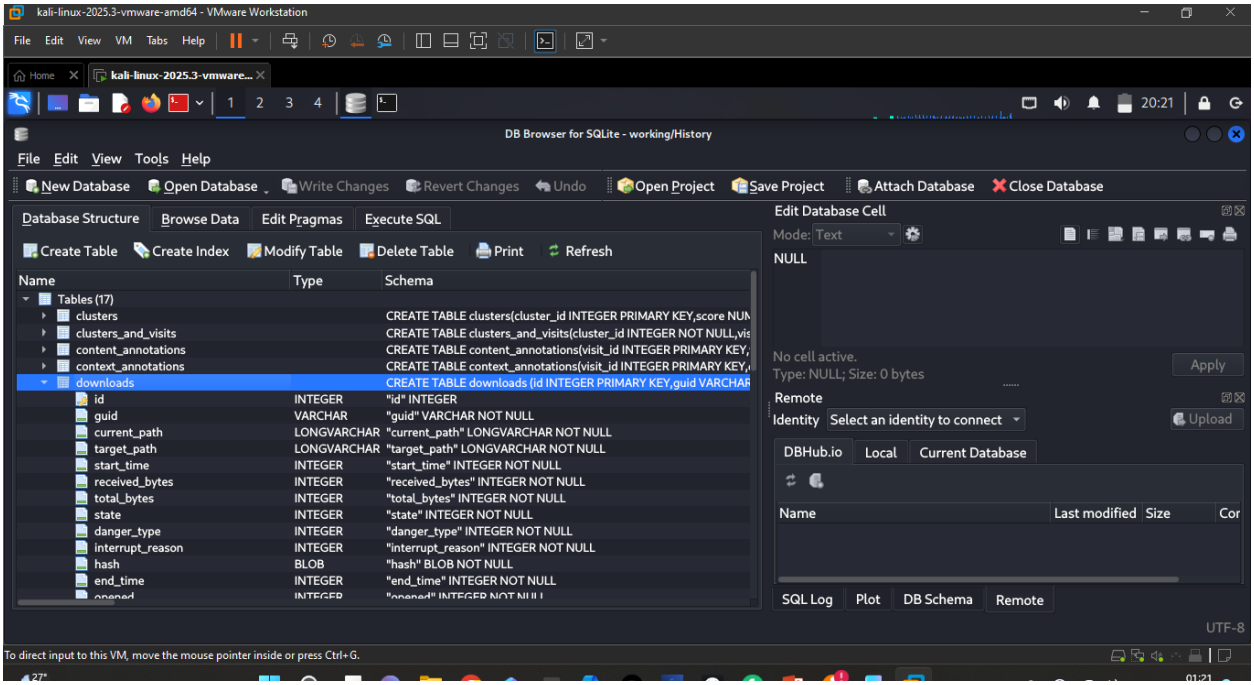



*Image 6 (E06). Showing the Downloads Structure which is to save location and other download details.*

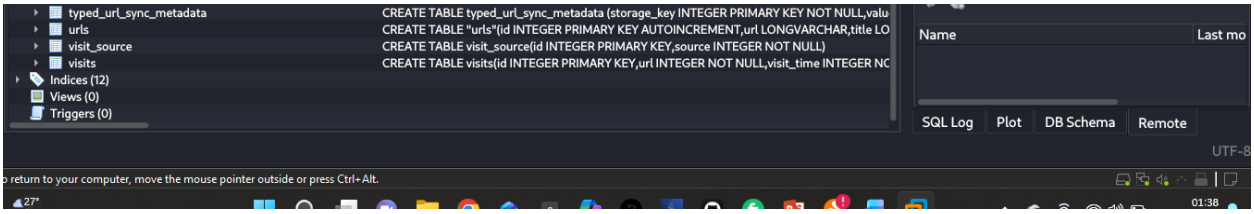



*Image 7 (E07). Showing the urls which Stores visited URLs, including page titles and when was the last time the URL was visited. (and) Visits structure that Stores each visit to a URL, including when the URL is visited and how long a user stay on the page.*

4. Building the Core Timeline

*Running the SQL Query (SELECT*
*v.id AS visit_id,*
*u.id AS url_id,*
*u.url,*
*u.title,*
*datetime(*
*v.visit_time/1000000 +*
*strftime('%s','1601-01-01'),*
*'unixepoch'*
*) AS visit_utc,*

*v.from_visit,*
*v.transition,*
*v.visit_duration/1000.0 AS duration_ms*
*FROM visits v*
*JOIN urls u*
*ON v.url=u.id*
*ORDER BY v.visit_time;) to have a primary timeline*

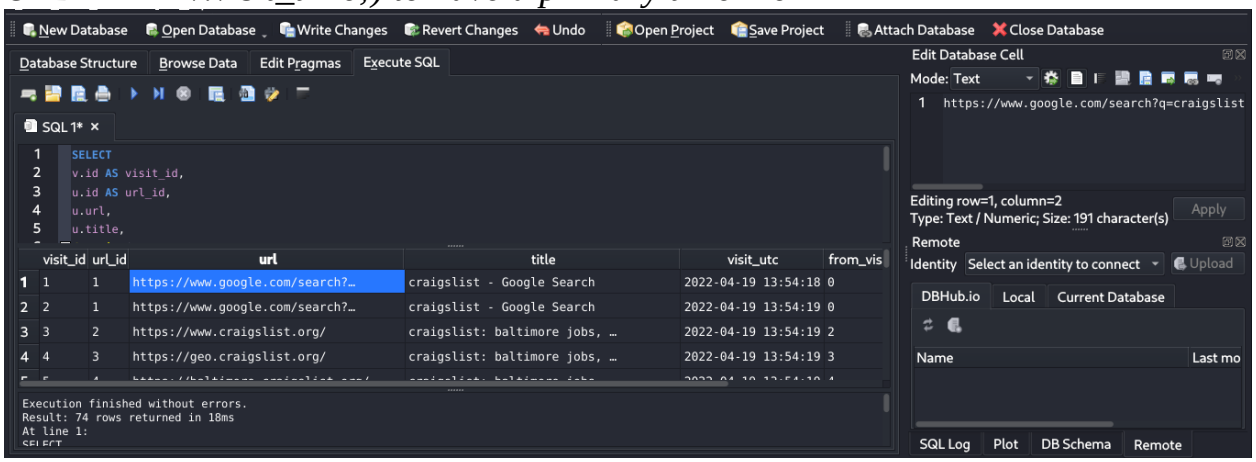



*Image 6 (E06) indicates the above query implemented with results of 74 rows returned in 18ms at line 1*

4.1Analysis Strategy

Segment visit records based on visit duration (Labeling each segment as 1,2,…..)

Seg 1: visit Id = 1-2 (“search Craiglist”)

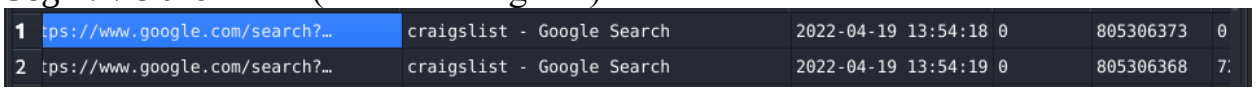
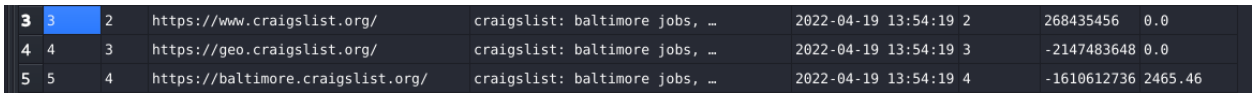



Image 7 (E07)

| Visit ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 1 | 2022-04-19 13:54:18 | User typed “Craiglist” into chrome address bar and generate a url | O seconds which indicate a redirect |
| 2 | 2022-04-19 13:54:19 | Without any delay (0 second duration), chrome sends the URL to Google and google returns results in web page with the title “Craiglist” – Google search | User stayed in the page for 7 seconds |

Seg 2 : visited ID = 3-5 (visited https://www.craigslist.org)

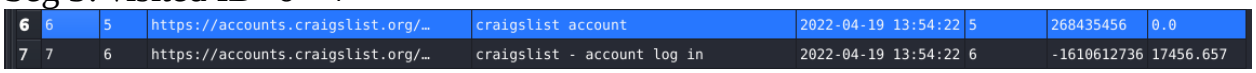


| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 3 | 3 | 2 | https://www.craigslist.org/ | craigslist: baltimore jobs, … | 2022-04-19 13:54:19 | 2 | 268435456 0.0 |
| 4 | 4 | 3 | https://geo.craigslist.org/ | craigslist: baltimore jobs, … | 2022-04-19 13:54:19 | 3 | -2147483648 0.0 |
| 5 | 5 | 4 | https://baltimore.craigslist.org/ | craigslist: baltimore jobs, … | 2022-04-19 13:54:19 | 4 | -1610612736 2465.46 |

Image 8 (E08)

| Visit ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 3 - 5 | 2022-04-19 13:54:19 | Suspect clicked the link (from visits id 2) and visited https://www.craiglist.org/. however, it was redirected to https://geo.craigslist.org/ and https://baltimore.craigslist.org/ | Suspect stayed in the page https://baltimore.craigslist.org/ for 2.46546 seconds |

Seg 3: visited ID=6 – 7



| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 6 | 6 | 5 | https://accounts.craigslist.org/… | craigslist account | 2022-04-19 13:54:22 | 5 | 268435456 0.0 |
| 7 | 7 | 6 | https://accounts.craigslist.org/… | craigslist - account log in | 2022-04-19 13:54:22 | 6 | -1610612736 17456.657 |

*Image 9 (E09)*

| Visit ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 6 - 7 | 2022-04-19 13:54:22 | The Suspect click and account link https://accounts.craiglist.org/login/home to login and was redirected to another link https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L | Suspect stayed in the page https://accounts.craigslist.org/login?rp=%2Flogin%2Fhome&rt=L for 17.456 seconds |

Seg 4: visited ID = 8

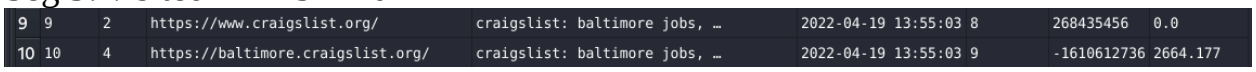


| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 8 | 8 | 5 | https://accounts.craigslist.org/… | craigslist account | 2022-04-19 13:54:39 | 7 | 805306375 23439.248 |

*Image 10 (E10)*

| Visit ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 8 | 2022-04-19 13:54:39 | The Suspect click and account link https://accounts.craigslist.org/login/home where the user filled out the login form and submit | Suspect stayed in the page 23 seconds |

Seg 5: visited ID = 9 – 10

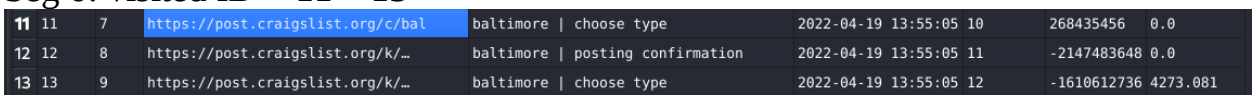


| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 9 | 9 | 2 | https://www.craigslist.org/ | craigslist: baltimore jobs, … | 2022-04-19 13:55:03 | 8 | 268435456 0.0 |
| 10 | 10 | 4 | https://baltimore.craigslist.org/ | craigslist: baltimore jobs, … | 2022-04-19 13:55:03 | 9 | -1610612736 2664.177 |

*Image 11 (E11)*

| Visit ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 9 - 10 | 2022-04-19 13:55:03 | The user clicked the link https://www.craigslist.org/ and was redirected to https://baltimore.craigslist.org/ | The user stayed on the page for 2.6 seconds |

Seg 6: visited ID = 11 – 13

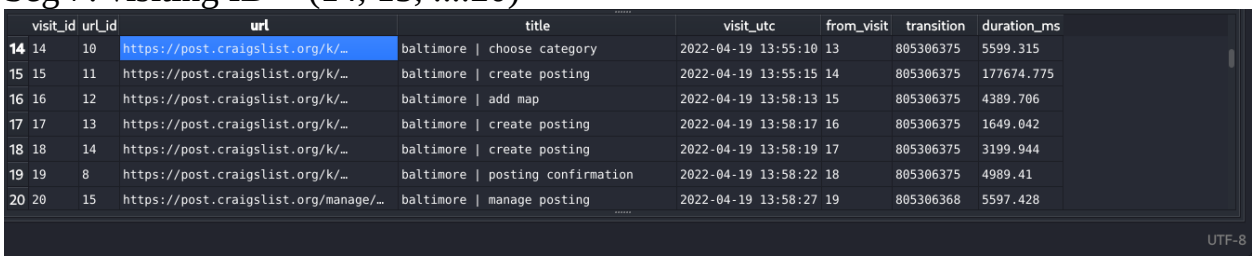


| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 11 | 11 | 7 | https://post.craigslist.org/c/bal | baltimore \| choose type | 2022-04-19 13:55:05 | 10 | 268435456 | 0.0 |
| 12 | 12 | 8 | https://post.craigslist.org/k/… | baltimore \| posting confirmation | 2022-04-19 13:55:05 | 11 | -2147483648 | 0.0 |
| 13 | 13 | 9 | https://post.craigslist.org/k/… | baltimore \| choose type | 2022-04-19 13:55:05 | 12 | -1610612736 | 4273.081 |

*Image 12 (E12)*

| Visit ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 11 - 13 | 2022-04-19 13:55:05 | The user clicked the link https://post.craigslist.org/c/bal and was redirected to https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5 User was also redirected to https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=type | The user stayed on the page for four (4) seconds |

Seg 7: visiting ID = (14, 15, ....20)



| | visit_id | url_id | url | title | visit_utc | from_visit | transition | duration_ms |
|---|---|---|---|---|---|---|---|---|
| 14 | 14 | 10 | https://post.craigslist.org/k/… | baltimore \| choose category | 2022-04-19 13:55:10 | 13 | 805306375 | 5599.315 |
| 15 | 15 | 11 | https://post.craigslist.org/k/… | baltimore \| create posting | 2022-04-19 13:55:15 | 14 | 805306375 | 177674.775 |
| 16 | 16 | 12 | https://post.craigslist.org/k/… | baltimore \| add map | 2022-04-19 13:58:13 | 15 | 805306375 | 4389.706 |
| 17 | 17 | 13 | https://post.craigslist.org/k/… | baltimore \| create posting | 2022-04-19 13:58:17 | 16 | 805306375 | 1649.042 |
| 18 | 18 | 14 | https://post.craigslist.org/k/… | baltimore \| create posting | 2022-04-19 13:58:19 | 17 | 805306375 | 3199.944 |
| 19 | 19 | 8 | https://post.craigslist.org/k/… | baltimore \| posting confirmation | 2022-04-19 13:58:22 | 18 | 805306375 | 4989.41 |
| 20 | 20 | 15 | https://post.craigslist.org/manage/… | baltimore \| manage posting | 2022-04-19 13:58:27 | 19 | 805306368 | 5597.428 |

UTF-8

*Image 13 (E13). Showing six (6) visited Id's with the urls, titles, utc, transitions and durations*

| Visit ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 14 | 2022-04-19 13:55:10 | The user clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=cat | The user stayed on the page for five (5) seconds |
| 15 | 2022-04-19 13:55:15 | The user clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=edit | The user stayed on the page for 177 seconds |
| 16 | 2022-04-19 13:58:13 | The user clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=geoverify | The user stayed on the page for 4 seconds |
| 17 | 2022-04-19 13:58:17 | The user clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=editimage | The user stayed on this page for 1.4 seconds |
| 18 | 2022-04-19 13:58:19 | The user under this ID clicked the link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5?s=preview | The user stays on the page for 3.1 seconds |
| 19 | 2022-04-19 13:58:22 | The user clicked this link https://post.craigslist.org/k/sHbOUOi_7BG68qUDIbeTbQ/bQTi5 | And stays on this page for approximately 5 seconds |
| 20 | 2022-04-19 13:58:27 | The user follows this link https://post.craigslist.org/manage/7473121658 | The user stays on the page for 5 seconds |

Seg 8: Visiting ID's = 21 – 23

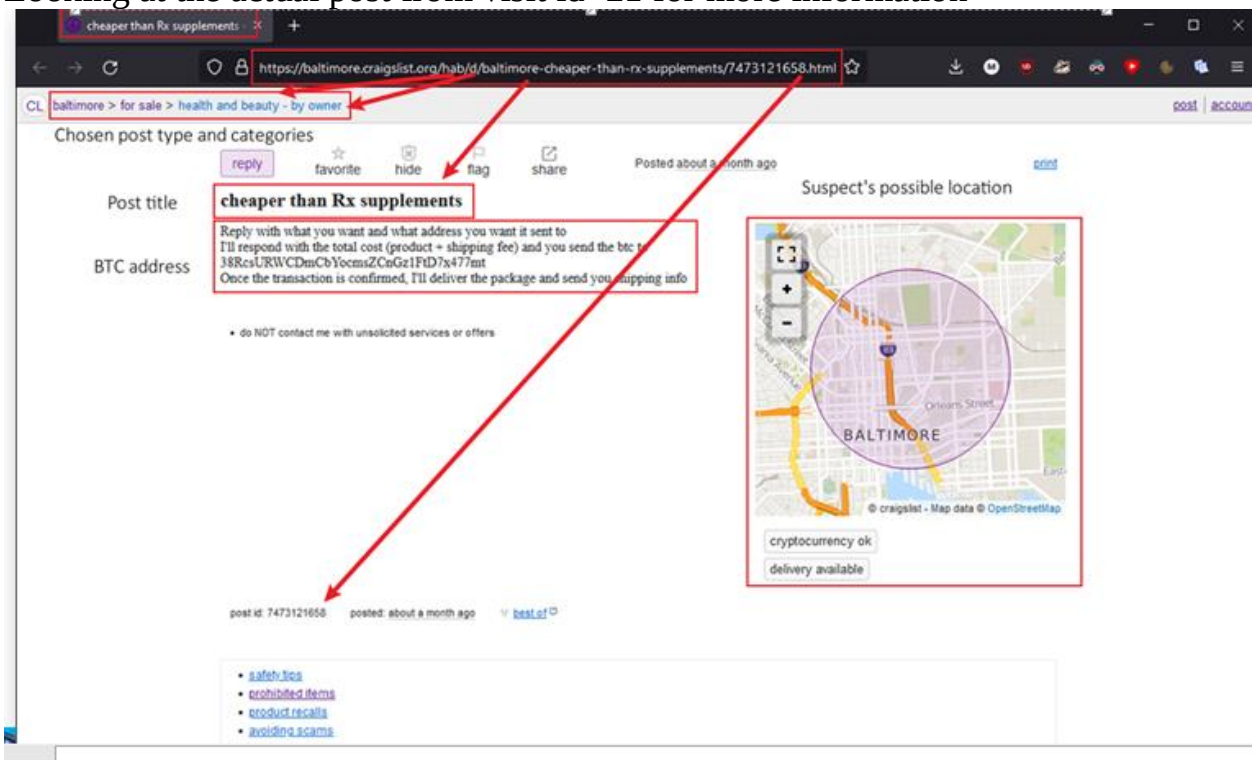
| | visit_id | url_id | url | title | visit_utc | from_visit | transition | duration_ms |
|---|---|---|---|---|---|---|---|---|
| 21 | 21 | 16 | https://baltimore.craigslist.org/hab… | cheaper than Rx supplements - health… | 2022-04-19 13:58:31 | 0 | 805306368 | 0.0 |
| 22 | 22 | 17 | https://www.google.com/search?… | gmail - Google Search | 2022-04-19 13:58:38 | 0 | 838860805 | 0.0 |
| 23 | 23 | 17 | https://www.google.com/search?… | gmail - Google Search | 2022-04-19 13:58:39 | 0 | 805306368 | 842.727 |

*Image 14 (E14)*

| Visit ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|

| | | | |
|---|---|---|---|
| 21 - 23 | 2022-04-19 13:58:31 | The user clicked the link https://baltimore.craigslist.org/hab/d/baltimore-cheaper-than-rx-supplements/7473121658.html) with path /hab/d/Baltimore-cheaper-than-rx-supplement to view post 747341212658<br>Then the user typed “Gmail” in chrome address bar and URL was generated to Google Search results | The user stayed on the page for 8. seconds |

Looking at the actual post from visit id=21 for more information




Seg 9: visiting ID’s = 24 – 26

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 24 | 24 | 18 | https://www.google.com/gmail/ | Gmail: Free, Private & Secure Email … | 2022-04-19 13:58:40 | 23 | 268435456 | 0.0 |
| 25 | 25 | 19 | https://mail.google.com/mail/ | Inbox (12) - unsub.fscs@gmail.com - … | 2022-04-19 13:58:40 | 24 | -2147483648 | 0.0 |
| 26 | 26 | 20 | https://mail.google.com/mail/u/0/ | Inbox (12) - unsub.fscs@gmail.com - … | 2022-04-19 13:58:40 | 25 | -2147483648 | 0.0 |

*Image 15 (E15)*

| Visited ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|

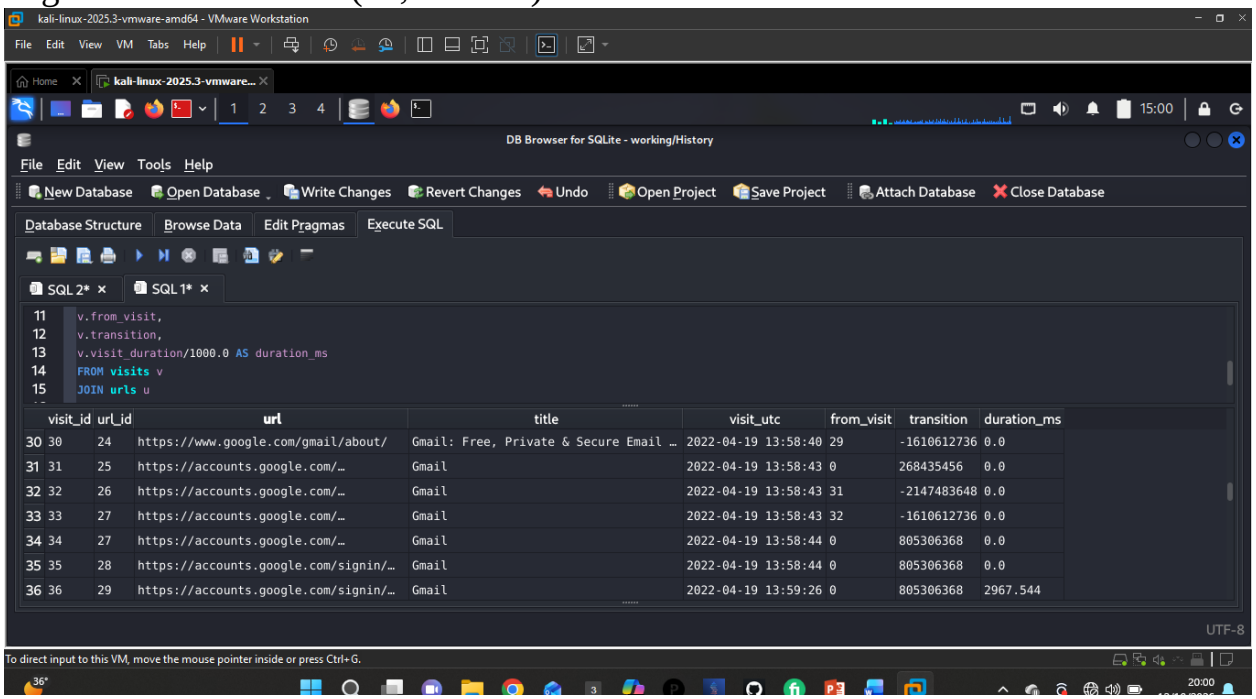
| | | | |
|---|---|---|---|
| 24 - 26 | 2022-04-19 13:58:40 | The user clicked the link https://www.google.com/gmail/ to Gmail then was redirectd to Inbox (12) - unsub.fscs@gmail.com - Gmail | The user stayed on the page for 0 seconds |

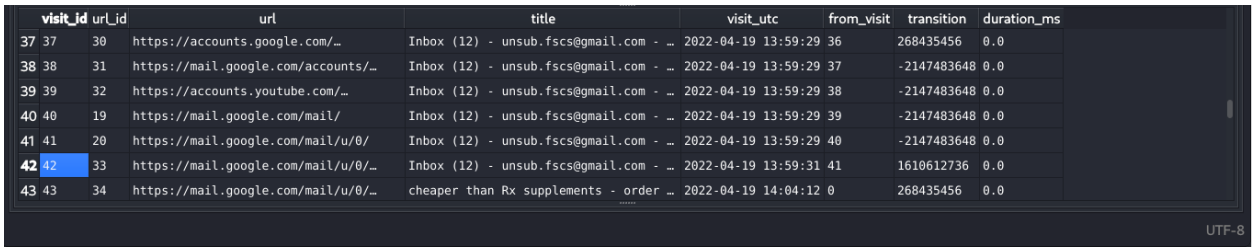
Seg 9: visited ID's = 27 – 29

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 27 | 27 | 21 | https://accounts.google.com/… | Gmail: Free, Private & Secure Email … | 2022-04-19 13:58:40 | 26 | -2147483648 | 0.0 |
| 28 | 28 | 22 | https://mail.google.com/intl/en/mail… | Gmail: Free, Private & Secure Email … | 2022-04-19 13:58:40 | 27 | -2147483648 | 0.0 |
| 29 | 29 | 23 | https://www.google.com/intl/en/mail/… | Gmail: Free, Private & Secure Email … | 2022-04-19 13:58:40 | 28 | -2147483648 | 0.0 |

| Visited ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 27 - 29 | 2022-04-19 13:58:40 | The user was redirected to accounts.google.com with the URL to their inbox as a parameter to ServiceLogin?service with https://mail.google.com/mall/u/O, User was then directed on ro an about.html page User was then redirected again to another sbout page with path /help/ to an about.html page | The user stayed on the page for 0 seconds |

Seg 10: Visited ID's = (30, 31....36)





| | visit_id | url_id | url | title | visit_utc | from_visit | transition | duration_ms |
|---|---|---|---|---|---|---|---|---|
| 37 | 37 | 30 | https://accounts.google.com/… | Inbox (12) - unsub.fscs@gmail.com - … | 2022-04-19 13:59:29 | 36 | 268435456 | 0.0 |
| 38 | 38 | 31 | https://mail.google.com/accounts/… | Inbox (12) - unsub.fscs@gmail.com - … | 2022-04-19 13:59:29 | 37 | -2147483648 | 0.0 |
| 39 | 39 | 32 | https://accounts.youtube.com/… | Inbox (12) - unsub.fscs@gmail.com - … | 2022-04-19 13:59:29 | 38 | -2147483648 | 0.0 |
| 40 | 40 | 19 | https://mail.google.com/mail/ | Inbox (12) - unsub.fscs@gmail.com - … | 2022-04-19 13:59:29 | 39 | -2147483648 | 0.0 |
| 41 | 41 | 20 | https://mail.google.com/mail/u/0/ | Inbox (12) - unsub.fscs@gmail.com - … | 2022-04-19 13:59:29 | 40 | -2147483648 | 0.0 |
| 42 | 42 | 33 | https://mail.google.com/mail/u/0/… | Inbox (12) - unsub.fscs@gmail.com - … | 2022-04-19 13:59:31 | 41 | 1610612736 | 0.0 |
| 43 | 43 | 34 | https://mail.google.com/mail/u/0/… | cheaper than Rx supplements - order … | 2022-04-19 14:04:12 | 0 | 268435456 | 0.0 |

UTF-8

*Image 16 (E16)*

| Visited ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 30 - 32 | 2022-04-19 13:58:40 | User clicked a link https://www.google.com/) withb path /gmail/about/ to Google Workspace then user click a link https://accounts.google.com/\| with path /accounts Chooser/ and query siginChooser?service=mail to a login page Then user was redirected to link (https://accounts.google.com) with query AccountChooser?service=email to choose an account | The user stayed on the page for 0 seconds |
| 33 - 36 | 2022-04-19 13:58:43 | The user clicks on the web on the url (https://account.google.com) gmail with query ServiceLogin?continue= to a login page<br>Then User clicked a link https://accounts.google.com/o to Gmail with same query (https://accounts.google.com) to Gmail with path /signin/ with query identifier?continue to a login page prompting identifier email address/username<br>Then user clicked a link (https://account.google.com/) with path /sigin/challenge/ and query pwd?continue= to a login page prompting challenge password | Time spend is on this pages is approximately 3 sconds |

| | | | |
|---|---|---|---|
| 37 - 39 | 2022-04-19 13:59:29 | User clicked the link (https://account.google.com) with query CheckCookie?continue= and parameter value loginDonetHtml to unsub.fscs@gmail.com's inbox<br>Then the user was redirected to (https://mail.google.com/) with path /accounts/ and query SetOSID?authuser to unsub.fscs@gmail.com's inbox and then the user was redirected (https://accounts.youtube.com/) with path /accounts/ and query SetSID? To unsub.fscs@gmail.com's inbox | The user stays on the page for 0 seconds |
| 40 - 41 | 2022-04-19 13:59:29 | User clicked a link (https://mail.google.com/) with path mail to unsub.fscs@gmail.com's inbox<br>User then click a link (https://mail.google.com) with path /u/o/ to unsub.fscs@gmail.com's inbox | The user stays on the page for 0 seconds |
| 42 - 43 | 2022-04-19 13:59:31 | User clicked on the url https://mail.google.com/mail/u/0/#inbox to unsub.fscs@gmail.com's inbox<br>User clicked a link https://mail.google.com/ with string WhctKKXXDrfCqXBGCnHkHGxZnZbk XKdXnJTmxwZwPtjVmvHMrfPPRthPXJ gXngHcFLmggWv. Indicating ID for opened email with subject header cheaper than Rx supplemnts - order | User stayed for 0 seconds on that page |

Seg 11: visiting ID's = (43, 44.....52)

| | visit_id | url_id | url | title | visit_utc | from_visit | transition | duration_ms |
|---|---|---|---|---|---|---|---|---|
| 44 | 44 | 35 | https://mail.google.com/mail/u/0/… | cheaper than Rx supplements - order … | 2022-04-19 14:04:35 | 43 | 1073741824 | 0.0 |
| 45 | 45 | 36 | https://mail.google.com/mail/u/0/… | cheaper than Rx supplements - order … | 2022-04-19 14:04:46 | 44 | 1073741824 | 0.0 |
| 46 | 46 | 34 | https://mail.google.com/mail/u/0/… | cheaper than Rx supplements - order … | 2022-04-19 14:04:52 | 45 | 1610612736 | 0.0 |
| 47 | 47 | 33 | https://mail.google.com/mail/u/0/… | Inbox (12) - unsub.fscs@gmail.com - … | 2022-04-19 14:05:59 | 0 | 805306368 | 0.0 |
| 48 | 48 | 37 | https://mail.google.com/mail/u/0/… | cheaper than Rx supplements -- txid:… | 2022-04-19 14:56:47 | 0 | 268435456 | 0.0 |
| 49 | 49 | 38 | https://www.google.com/url?q=https:/… | Imgur: The magic of the Internet | 2022-04-19 14:56:52 | 0 | 268435456 | 0.0 |
| 50 | 50 | 39 | https://imgur.com/dTgrkP7 | Imgur: The magic of the Internet | 2022-04-19 14:56:52 | 49 | -1610612736 | 18926.083 |
| 51 | 51 | 40 | https://www.google.com/search?… | blockchain explorer - Google Search | 2022-04-19 14:58:59 | 0 | 838860805 | 0.0 |
| 52 | 52 | 40 | https://www.google.com/search?… | blockchain explorer - Google Search | 2022-04-19 14:59:00 | 0 | 805306368 | 1026.231 |

UTF-8

*Image 17 (E17)*

| Visited ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 44 - 45 | 2022-04-19 14:04:35 | The user clicked the link (https://www.google.com/gmail/) with the query ?compose=JHrtff referring to open message WhtcKKXX with subject header "order"<br>User then clicked a link (https://mail.google.com) with query ?compose=DXDWs referring to same message WhcKKXX indicating two replies were made for the same message | The user stayed on the page for 0 seconds |
| 46 - 48 | 2022-04-19 14:04:52 | User clicked on the link https://mail.google.com/mail/u/0/#inbox/ with the strings WhctKKXXDrfCqXBGC "order" message open<br>User click on the link https://mail.google.com/mail/u/0/#inbox/ with strings WhctKKXXDrfCqvtRj, another new message with subject header "txid: 517b2159….." | User datyed in the page for 0 seconds |
| 49 - 50 | 2022-04-19 14:56:52 | User clicked a link (https://www.google.com/) with query url?=https://imgur.com/dtgrkp7 and parameter source=gmail indicating the imgur link originated from Gmail message<br>User clicked a link (https://imgur.com/dtgrkp7) to imgur post | User stayed on the page for 18 seconds |
| 50 | | Looking at the imgur post – URL does not have enough information about the image post<br>Message shows Bitcoin was sent<br>Amount of BTC sent and USD equivalent at time<br>Suspect's wallet address from craigslist post (38RcsURWCDmCBYocmsZCnGz1FtD7x477mt)<br>And another link for block chain explorer | |

| | | | |
|---|---|---|---|
| | | (E18) below shows more details. | |

*Imgur is an online image sharing and hosting service. Images can be uploaded anonymously and shared. In database, you can see the imgur link from message has parameter source=gmail*

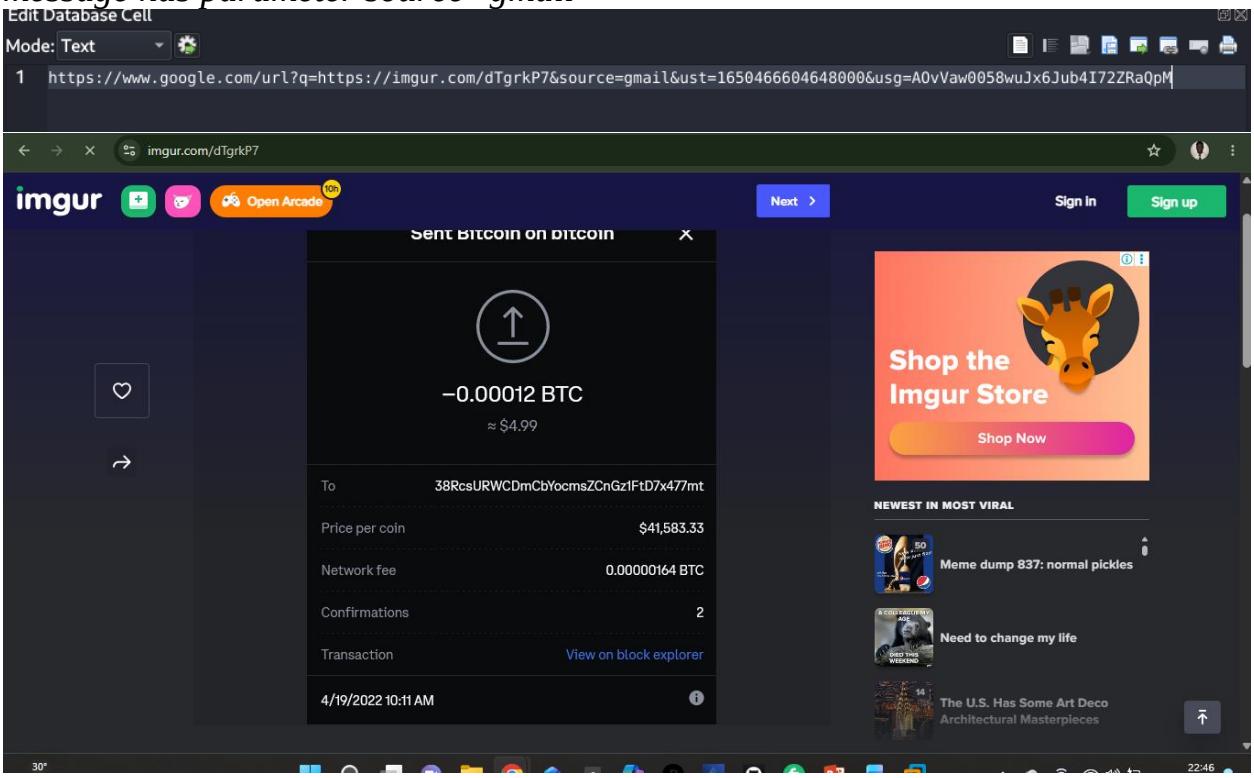



*Image 18 (E18).*

*Blockchain Explorer is a block and crypto transaction search engine – queries can include wallet addresses and transaction IDs*

Using the TXID from the Gmail message we can search it on a Blockchain Explorer to verify authenticity

Seg 12: Visiting ID's 51 – 59

| | visit_id | url_id | url | title | visit_utc | from_visit | transition | duration_ms |
|---|---|---|---|---|---|---|---|---|
| 51 | 51 | 40 | https://www.google.com/search?… | blockchain explorer - Google Search | 2022-04-19 14:58:59 | 0 | 838860805 | 0.0 |
| 52 | 52 | 40 | https://www.google.com/search?… | blockchain explorer - Google Search | 2022-04-19 14:59:00 | 0 | 805306368 | 1026.231 |
| 53 | 53 | 41 | https://www.blockchain.com/explorer | Blockchain Explorer - Search the … | 2022-04-19 14:59:01 | 52 | 805306368 | 0.0 |
| 54 | 54 | 41 | https://www.blockchain.com/explorer | Blockchain Explorer - Search the … | 2022-04-19 14:59:02 | 0 | 805306368 | 18679.186 |
| 55 | 55 | 42 | https://www.blockchain.com/search?… | Transaction: … | 2022-04-19 14:59:06 | 54 | 268435456 | 0.0 |
| 56 | 56 | 43 | https://www.blockchain.com/btc/tx/… | Transaction: … | 2022-04-19 14:59:06 | 55 | -1610612736 | 0.0 |
| 57 | 57 | 43 | https://www.blockchain.com/btc/tx/… | Transaction: … | 2022-04-19 14:59:06 | 54 | 805306368 | 0.0 |
| 58 | 58 | 44 | https://www.blockchain.com/btc/… | Address: … | 2022-04-19 14:59:21 | 54 | 805306368 | 0.0 |

*Image 19 (E19)*

| Visited ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 51 - 52 | 2022-04-19 14:58:59 | The user typed “Blockchain Explorer” in the search bar and generated a url to google search result | The user stayed on the page for 1 seconds |
| 53 - 54 | 2022-04-19 14:59:01 | User clicked a link (https://www.blockchain.com/explorer) to Blockchain Explorer | The user stayed for 18 seconds |
| 55 - 57 | 2022-04-19 14:59:06 | User clicklink (https://www.blockchain) with query ?search=value 517b215….<br>Then a Url was generated<br>User then clicked a link (https://blockchain.com) with path /btc/tx and string matching 517b215 transaction to view the transaction | The user stayed in the page for 0 seconds |
| 58 | 2022-04-19 14:59:21 | User clicked a url of blockchain.com with path /btc/address/ and string 388csURW | User stayed for 0 seconds |
| 59 - 60 | 2022-04-19 14:59:38 | User insert “Kraken” into chrome search bar and generated a URL to Google Search results | User stayed for 7 seconds |

*Suspect clicked on their wallet address from transaction page and transition to address page*

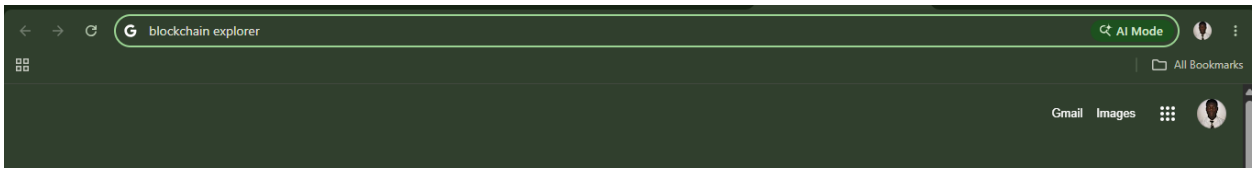



*Image 20 (19)*

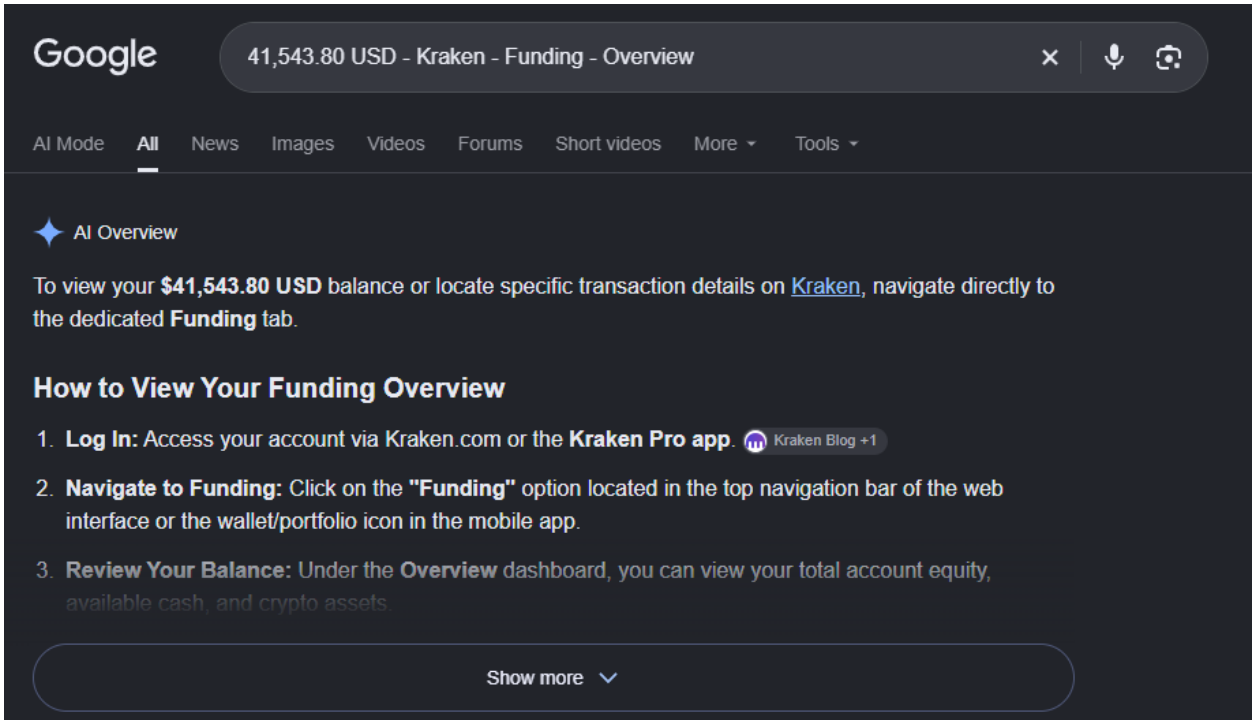
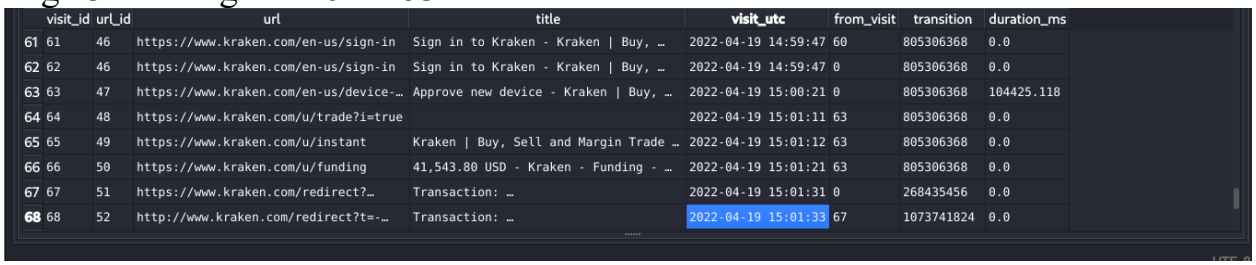



*Image 21 (20)*
*Kraken is a cryptocurrency exchange and banking platform*

Seg 13: visiting ID's 61 – 68



| | visit_id | url_id | url | title | visit_utc | from_visit | transition | duration_ms |
|---|---|---|---|---|---|---|---|---|
| 61 | 61 | 46 | https://www.kraken.com/en-us/sign-in | Sign in to Kraken - Kraken \| Buy, … | 2022-04-19 14:59:47 | 60 | 805306368 | 0.0 |
| 62 | 62 | 46 | https://www.kraken.com/en-us/sign-in | Sign in to Kraken - Kraken \| Buy, … | 2022-04-19 14:59:47 | 0 | 805306368 | 0.0 |
| 63 | 63 | 47 | https://www.kraken.com/en-us/device-… | Approve new device - Kraken \| Buy, … | 2022-04-19 15:00:21 | 0 | 805306368 | 104425.118 |
| 64 | 64 | 48 | https://www.kraken.com/u/trade?i=true | | 2022-04-19 15:01:11 | 63 | 805306368 | 0.0 |
| 65 | 65 | 49 | https://www.kraken.com/u/instant | Kraken \| Buy, Sell and Margin Trade … | 2022-04-19 15:01:12 | 63 | 805306368 | 0.0 |
| 66 | 66 | 50 | https://www.kraken.com/u/funding | 41,543.80 USD - Kraken - Funding - … | 2022-04-19 15:01:21 | 63 | 805306368 | 0.0 |
| 67 | 67 | 51 | https://www.kraken.com/redirect?… | Transaction: … | 2022-04-19 15:01:31 | 0 | 268435456 | 0.0 |
| 68 | 68 | 52 | http://www.kraken.com/redirect?t=-… | Transaction: … | 2022-04-19 15:01:33 | 67 | 1073741824 | 0.0 |

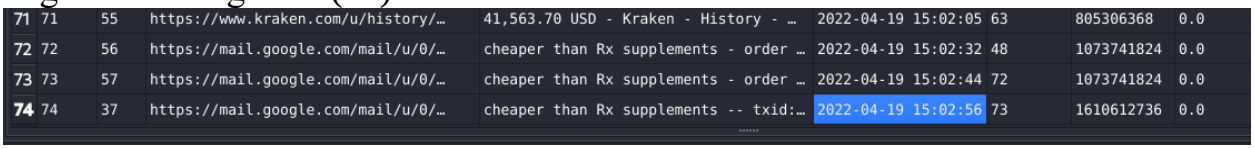
*Image 21 (E21)*

| Visited ID | Url ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|---|
| 61 - 62 | 46 | 2022-04-19 14:59:47 | User clicked a link (https://kraken.com) with path /en-us/sign-in to Kraken Sign to page | The user stayed on the page for 0 seconds |
| 63 - 65 | 47,48,49 | 2022-04-19 15:00:21 | User clicked a link (https://kraken.com) with path /en-us/device-approval to approve signing in from new device | User stayed for 104 seconds |

| | | | | |
|---|---|---|---|---|
| | | | User then clicked a link (https://www.karaken.com) with path /u/ and query trade?i=true after successfully signing in<br>User then clicked a link (https://www.karaken.com) with path /u/instant | |
| 66 | 50 | 2022-04-19 15:01:21 | User clicked a link (https://www.kraken.com) with path /u/funding to<br>Funding pae | User stays for 0 seconds |
| 67 - 70 | 51, 52, 53 and 54 | 2022-04-19 15:01:31 | User clicked link (https;//www.kraken.com) with query redirect?url= pointing to https://mempool.space/tx/517b21...<br>Then page is refreshed by server and redirected to link (https://www.kraken.com) with query redirect? And title Transaction: 517b21…<br>Then redirected to link (https://mempoolspace) with path /tx/517b21… | User stays for 0 seconds |

*(ID's 63 – 65). This segment are transitions for suspect authenticating device login and successfully logging in*
*mempool space is like Blockchain Explorer, allowing search of BTC transaction, address history, and transaction status*

Seg 14: Visiting ID's (71)



| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 71 | 71 | 55 | https://www.kraken.com/u/history/… | 41,563.70 USD - Kraken - History - … | 2022-04-19 15:02:05 | 63 | 805306368 | 0.0 |
| 72 | 72 | 56 | https://mail.google.com/mail/u/0/… | cheaper than Rx supplements - order … | 2022-04-19 15:02:32 | 48 | 1073741824 | 0.0 |
| 73 | 73 | 57 | https://mail.google.com/mail/u/0/… | cheaper than Rx supplements - order … | 2022-04-19 15:02:44 | 72 | 1073741824 | 0.0 |
| 74 | 74 | 37 | https://mail.google.com/mail/u/0/… | cheaper than Rx supplements -- txid:… | 2022-04-19 15:02:56 | 73 | 1610612736 | 0.0 |

*Image 22 (E21)*

| Visited ID | Date & Time | Description | Duration (in seconds) |
|---|---|---|---|
| 71 | 2022-04-19 15:02:05 | User clicked a link (https://www.kraken.com) with path | The user stayed on the |

| | | | |
|---|---|---|---|
| | | /u/history/ledger to Kraken History Ledger page | page for 0 seconds |
| 72 - 73 | 2022-04-19 15:02:32 | User clicked a link (https://mail.google.com) with path /u/0/#inbox with string pointing to WhctKKXXdrfC… message with subject “order” and query ?compose=JhetffdgkF.. as one reply to the message<br>User then clicked a link (https://mail.google.com) with same path pointing to same WhctKKXXdrfC… message but with another reply xQTrpQHKf…. | The user stayed on the page for 0 seconds |
| 74 | 2022-04-19 15:02:56 | User clicked a link (https://mail.google.com) with path /mail/u/o/#inbox/ with string WhctKKXXDRf message with subject header txid from earlier visit id=48 | The user stayed on the page for 0 seconds |

Limitations

1. Browser history does not prove who used the device.
2. History records may be incomplete or deleted.